Exponential & Logarithmic Functions: Compound Interest & LN

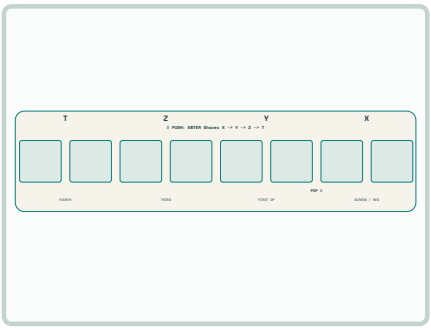
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.HSF.LE.A.4

Student Name: _____ Date: _____ **Model: SC-32** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-32 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Stack Board

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Represent exponential doubling by stacking power-of-two tiles (1, 2, 4, 8, 16, 32).

Sketch Model on Paper:
Draw exponential growth curve showing rapid upward curvature.

3. Step-by-Step LCD Stack Trace (SC-32)

MODE:	123	Base-10 Arithmetic	
1.	1.05	X: 1.05	Growth factor 1.05 i
2.	ENTER	Y: 1.05 | X: 1.05	Stack lift
3.	10	Y: 1.05 | X: 10	Exponent t = 10 into
4.	y^x	X: 1.6289	Power: 1.05^{10} = 1.62

EXPECTED LCD X: $1,628.89

1 **Calculate $600 at 5% for 4 years: 600 × $(1.05)^4$.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-32

2 **Calculate $3,000 at 3% for 10 years: 3000 × $(1.03)^{10}$.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-32

3 **Evaluate natural log ln(50) on SC-32: [50] [LN].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-32

4 **Evaluate log base 10 of 1,000 on SC-32: [1000] [LOG].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-32

Station Diagnostic & Solution Key: Chapter 6

Exponential & Logarithmic Functions: Compound Interest & LN • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Stack Stage Rail & Token Tiles

CAD Blueprint: stack_board

Target Device: SC-32 Scientific

CAD / STL Source File:

Education/build/scad/stack_board.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal 4-Register Rail & White Tiles (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Calculate $600 at 5% for 4 years: 600 × (1.0** Standard: CCSS.MATH.HSF.LE.A.4	$729.30	[1][.][0][5] [ENTER] [4] [y^x] [6][0][
2	**Calculate $3,000 at 3% for 10 years: 3000 ×** Standard: CCSS.MATH.HSF.LE.A.4	$4,031.75	[1][.][0][3] [ENTER] [1][0] [y^x] [3][
3	**Evaluate natural log ln(50) on SC-32: [50] [** Standard: CCSS.MATH.HSF.LE.A.4	3.9120	[5][0] [LN]
4	**Evaluate log base 10 of 1,000 on SC-32: [100** Standard: CCSS.MATH.HSF.LE.A.4	3.0000	[1][0][0][0] [LOG]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Multiply by principal P AFTER computing the exponential power (1+r)^t, not before.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-32, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.